

# Z-basis metrics for parameterized quantum circuits: exact identities, structural blind spots, pole-damped optimization, and a relaxation-aware entropy benchmark

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## Abstract

The Qang ( $qg$ ) framework [1] expresses single-qubit rotations through two measurement-level quantities, the polar bias  $qg_Z = \langle \sigma_z \rangle$  and the outcome entropy  $qg_S = H(p_0)$ . We report results obtained with the open-source reference implementation **qang**. First, the two metrics are linked by an exact, branch-free identity,  $qg_S = H((1 + qg_Z)/2)$ , which holds for any single-qubit state. Consequently, any new multi-qubit information must come from the joint outcome distribution. We name the resulting Z-basis total correlation and give its bounds and its scope: it is not an entanglement measure. Second, we identify exact blind spots: graph-state entanglement and readout-only dephasing are invisible to every Z-diagonal statistic, and qg-space updates cannot reach optima outside  $[0, \pi]$  because of the arccos range. On  $H_2$  they recover none of the correlation energy. Third, we study a pole-damped gradient step whose damping schedule is  $|dqg_Z/d\theta|$ . On a 300-landscape benchmark,  $H_2$ , and a 4-qubit LiH Hamiltonian with a mixed  $R_y/R_x$  ansatz, it trades 3–20 $\times$  more iterations at well-tuned learning rates for convergence where plain gradient descent diverges. Fourth, as a benchmark,  $qg_S$  tracks heavy output probability ( $r = -0.93$ ) and needs no calibration, unlike linear XEB. However, it is *not* monotonic under amplitude damping: under device-calibrated noise with added idle relaxation it falls below its own noiseless value. Pairing  $qg_S$  with the register-averaged  $qg_Z$  removes the ambiguity in 23 of 24 circuits tested. A summary section lists twenty-six follow-up studies (QML encoding, control quantization, noise-type detection, circuit knitting, chemistry, error mitigation, thermal states, error correction, Hubbard dynamics, constrained QAOA, qubit characterization, syndrome tracking, coherence, few-shot estimation, IonQ noise models and shot-noise propagation in registers), each with its classical or standard control; the symmetry-witness results are developed in a companion paper. Every number is reproduced by a script and pinned by a regression test in the public repository.

## 1 Introduction

Parameterized quantum circuits are written in angles, but everything a device returns is a distribution over measurement outcomes. The Qang framework [1] makes that second view the primary one: for a qubit at polar angle  $\theta$ ,

$$qg_Z(\theta) = \cos \theta = \langle \sigma_z \rangle \in [-1, 1], \quad qg_S(\theta) = H(\cos^2 \frac{\theta}{2}) \in [0, 1], \quad (1)$$

with  $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ . This paper collects results obtained with the reference implementation **qang**, with the aim of stating where the metrics are exact, where they are blind, and where they mislead. Section 2 gives exact identities, Section 3 the structural blind spots,

Section 4 a pole-damped optimizer on molecular Hamiltonians, and Section 5 the behavior of  $qg_S$  as a benchmark, including a device-calibrated validation. Section 6 summarizes later results, and Section 7 lists the limitations together.

All results are produced by scripts in the `examples/` folder of the repository and pinned by regression tests (Section 8). Molecular Hamiltonians were derived once with PySCF [2] and are hard-coded, so neither PySCF nor qiskit-nature is a dependency.

## 2 Exact identities

### 2.1 A branch-free link between $qg_Z$ and $qg_S$

Inverting  $qg_S \mapsto \theta$  requires choosing a half-domain, because  $qg_S(\theta) = qg_S(\pi - \theta)$ . The opposite direction needs no choice.

**Proposition 1.** *For any single-qubit density matrix  $\rho$ , with  $qg_Z = \text{Tr}(\rho\sigma_z)$  and  $qg_S = H(\langle 0|\rho|0\rangle)$ ,*

$$qg_S = H\left(\frac{1 + qg_Z}{2}\right), \quad \frac{d qg_S}{d qg_Z} = \frac{1}{2} \log_2 \frac{1 - qg_Z}{1 + qg_Z}. \quad (2)$$

*Proof.*  $qg_Z = p_0 - p_1 = 2p_0 - 1$  with  $p_0 = \langle 0|\rho|0\rangle$ ; substitute into  $H(p_0)$  and differentiate.  $\square$

The proof uses only the Z-diagonal population. The identity therefore holds however the state was prepared ( $R_y$ ,  $R_x$ , a noisy channel, or a partial trace of a larger register), and the tests check it on random Haar states. The derivative diverges at  $qg_Z = \pm 1$ . This is the entropy-side counterpart of the Jacobian singularity  $d\theta/dqg_Z = -1/\sin\theta$ .

### 2.2 Z-basis total correlation

Applied to each reduced single-qubit state, Proposition 1 makes the per-qubit  $qg_Z$  profile and the per-qubit marginal  $qg_S$  two encodings of the same information. Anything new about an  $n$ -qubit register must come from the joint distribution of the Z-basis outcomes  $X_1, \dots, X_n$ . We define

$$C_Z(\rho) = \sum_{i=1}^n H(X_i) - H(X_1, \dots, X_n), \quad (3)$$

Watanabe’s total correlation [3] of the outcomes. For  $n = 2$  it equals the mutual information. By subadditivity,  $0 \leq C_Z \leq n - 1$  bits, with  $C_Z = 0$  iff the outcomes are independent. Product states give 0, Bell states 1 and GHZ $_n$  states  $n - 1$  (the maximum).

**Remark 1** (Scope).  $C_Z$  is a classical correlation measure and is not a witness of entanglement in either direction. The separable mixture  $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$  has  $C_Z = 1$ , as does a Bell state. Every graph state has  $C_Z = 0$  (Section 3).

## 3 Structural blind spots

**Graph states.** A graph state [4] is  $H^{\otimes n}$  followed by CZ gates.  $H^{\otimes n}$  makes every  $|\text{amplitude}|^2$  equal to  $2^{-n}$ , and CZ is diagonal. For every graph, therefore, each qubit has  $qg_Z = 0$ , the joint  $qg_S$  is maximal, and  $C_Z = 0$ . The entanglement, certified independently through the stabilizers  $X_i \prod_{j \in N(i)} Z_j$ , lives entirely in phases.

**Readout dephasing.** Dephasing preserves the diagonal of  $\rho$ . Applied only before measurement, it therefore leaves  $qg_S$  exactly unchanged at every strength, including full dephasing (checked to machine precision on a 4-qubit Quantum Volume circuit). The same channel applied *inside* the circuit is visible: later entangling gates turn the lost coherence into randomized populations, and  $qg_S$  rises monotonically.

Table 1: Success rates over 300 random one-qubit landscapes per learning rate, started near a pole.

| $\eta$         | 0.3   | 1.0   | 2.0   | 3.0   | 5.0   | 10.0  |
|----------------|-------|-------|-------|-------|-------|-------|
| plain success  | 0.997 | 0.773 | 0.187 | 0.087 | 0.057 | 0.027 |
| damped success | 0.963 | 1.000 | 0.477 | 0.357 | 0.253 | 0.130 |
| damped trapped | 0.017 | 0.000 | 0.030 | 0.023 | 0.040 | 0.067 |

**The arccos range.** A qg-space parameter update is mapped back through  $\theta = \arccos(qg_Z) \in [0, \pi]$ . On the  $\text{H}_2$  Hamiltonian (0.735 Å, STO-3G, two qubits after parity reduction), the single-excitation optimum lies at  $\theta \approx -0.22$ . Starting from Hartree–Fock, raw, clipped and Tikhonov-regularized qg-space descent all stop at exactly the Hartree–Fock energy ( $-1.1169990$  Ha, error  $2.0 \times 10^{-2}$  Ha). They recover none of the correlation energy, whereas  $\theta$ -space descent reaches the FCI energy to  $10^{-9}$  Ha. Regularizing the Jacobian cannot fix this, because the limit comes from the range of arccos, not from its derivative.

## 4 Pole-damped gradient descent

To keep updates in  $\theta$ -space while still using the pole geometry, we damp each parameter’s step by  $|dqg_Z/d\theta_i|$ , floored at  $\varepsilon$ :

$$\theta_i \leftarrow \theta_i - \eta \max(|\sin \theta_i|, \varepsilon) \frac{\partial E}{\partial \theta_i}. \quad (4)$$

This is position-dependent trust-region damping in the Levenberg–Marquardt tradition [5, 6]. It is not a new class of optimizer; the only framework-specific element is that the schedule is derived from the geometry rather than tuned. Because the factor depends only on populations, and  $R_x(\theta)|0\rangle$  and  $R_y(\theta)|0\rangle$  have identical populations, it applies unchanged to either axis. We use  $\varepsilon = 0.05$  throughout; results are insensitive to this choice.

**Random landscapes.** For 300 random landscapes  $E = h_z \cos \theta + h_x \sin \theta$  per learning rate, started near a pole (Table 1), damping costs a few percent of success at a safe rate. At  $\eta = 1$  it succeeds in 100% of cases versus 77%, and at larger rates it succeeds 2.5–5 times more often. The cost is a trapping mode, in which the optimizer oscillates at its starting pole; it reaches 6.7% at  $\eta = 10$ .

**$\text{H}_2$ , two coupled parameters.** Ansatz  $R_y(\theta_0) \otimes R_y(\theta_1)$  followed by CX; the optimum has  $\theta_0 = \pi$  (a pole). From  $(1.0, 10^{-6})$ , damping needs about  $3\times$  more steps to reach chemical accuracy at safe rates (603 vs. 196 at  $\eta = 0.05$ ). At  $\eta = 2.5, 3$  and 4, plain descent never reaches chemical accuracy, while the damped version reaches the FCI energy in 10, 8 and 6 steps.

**LiH, mixed axes.** LiH at 1.5459 Å (STO-3G, 2 electrons in 3 spatial orbitals, parity mapping) gives 4 qubits and 52 Pauli terms, the scale of early hardware VQE work [7, 8]. The ansatz is  $R_y(\theta_0)$ ,  $R_y(\theta_1)$  on the occupied orbitals and  $R_y(\theta_2)$ ,  $R_x(\theta_3)$  on the virtual ones, followed by CX( $2 \rightarrow 0$ ) and CX( $3 \rightarrow 1$ ). At  $(\pi, \pi, 0, 0)$  its state has overlap 1 with qiskit-nature’s Hartree–Fock circuit. All four optimal parameters lie within 0.04 rad of a pole, which makes the cost larger than for  $\text{H}_2$ : 355 vs. 17 steps at  $\eta = 0.3$ . The benefit is also larger (Table 2). The  $R_x$  phase is physical: after the CX it changes the energy relative to  $R_y$  at the same angle, although the single-qubit populations are identical.

Table 2: LiH, start  $(\pi, \pi, 10^{-6}, 10^{-6})$ : steps to sustained convergence ( $10^{-5}$  Ha) to the ansatz optimum  $-0.0440152$  Ha (active-space electronic energy; HF  $-0.0437813$ , FCI  $-0.0448309$ ).

| $\eta$         | 0.3 | 1.0 | 2.0   | 2.5   | 4.0   | 7.0   | 10.0  |
|----------------|-----|-----|-------|-------|-------|-------|-------|
| plain $\theta$ | 17  | 5   | never | never | never | never | never |
| pole-damped    | 355 | 106 | 53    | 42    | 26    | 15    | 10    |

## 5 $qg_S$ as a benchmark

**Heavy output probability and XEB.** On 10 random 4-qubit Quantum Volume circuits [9] mixed with global depolarizing noise,  $qg_S$  and heavy output probability have Pearson  $r = -0.926$  over 110 points. Linear XEB [10] has a noiseless value  $A - 1$ , with  $A = 2^n \sum_x p(x)^2$  ranging from 1.23 to 2.33 on the same circuits family ( $n = 3-6$ ), so it needs circuit-specific calibration;  $qg_S$  does not. Conversely, XEB’s estimator is a sample mean and is unbiased at any shot count, while the plug-in entropy is biased low (bias  $-0.054$  at 50 shots on 4 qubits). The Miller-Madow correction [11] reduces that bias about fivefold.

**Amplitude damping breaks monotonicity.** The reading “more noise  $\Rightarrow$  higher  $qg_S$ ” holds only for channels whose fixed point is maximally mixed. With amplitude damping applied after every two-qubit block (Fig. 1a),  $qg_S$  rises from 0.919 to 0.983 at  $\gamma = 0.1$  and then falls to 0 at  $\gamma = 1$ . At  $\gamma = 0.4$  it reads 0.918, almost the noiseless value.

**The fix: register-averaged  $qg_Z$ .** The mean  $q\bar{g}_Z = \frac{1}{n} \sum_i \text{Tr}(\rho_i \sigma_z)$  is pulled towards 0 by unital noise and towards +1 by relaxation. At  $\gamma = 0$  and 0.4 the pairs  $(qg_S, q\bar{g}_Z)$  are (0.919,  $-0.029$ ) and (0.918,  $+0.310$ ).  $q\bar{g}_Z$  rose monotonically along a 21-point  $\gamma$  grid in 23 of 24 circuits ( $n = 3, 4, 5$ ; seeds 0–7); the exception dips by 0.015 before rising.  $q\bar{g}_Z$  is linear in  $\rho$ , so its estimator from counts is an exact sample mean (Appendix A).

**Device-calibrated validation.** We ran the same circuit through Qiskit Runtime on the `fake_brisbane` backend, a simulator whose noise model is built from a real IBM device’s calibration data. An idle delay before readout served as a T1 knob (Fig. 1b). Heavy output probability fell monotonically (0.685 to 0.448) and  $q\bar{g}_Z$  rose monotonically ( $-0.036$  to  $+0.370$ ).  $qg_S$  rose to 0.977 at 50  $\mu\text{s}$  and then fell to 0.892 at 200  $\mu\text{s}$ , below its noiseless value of 0.919. Read on its own,  $qg_S$  would rank the most relaxed operating point as the least noisy.

The same script evaluated the LiH ansatz without error mitigation. The raw error was  $+0.036$  Ha at Hartree-Fock and  $+0.035$  Ha at the optimum. This is about  $20\times$  chemical accuracy and about  $150\times$  the 0.23 mHa difference the optimizer resolves noiselessly. A hardware run of this VQE therefore needs mitigation first.

**Randomized benchmarking and ZNE.** Under per-gate depolarizing noise  $p$ , the RB survival signal is exactly  $qg_Z(m) = (1 - p)^{m+1}$ , identical across random Clifford sequences to 10 digits in our runs [12]. For zero-noise extrapolation [13], extrapolation in  $qg_Z$ -space is exact for Bloch-vector shrinkage but biased ( $1.8 \times 10^{-3}$ ) for coherent angle drift;  $\theta$ -space shows the reverse. One should extrapolate in whichever space makes the noise linear. In barren-plateau settings [14, 15], regularized  $qg$ -space gradients keep the exponential variance decay, and raw  $qg$ -space gradients still diverge at the poles.

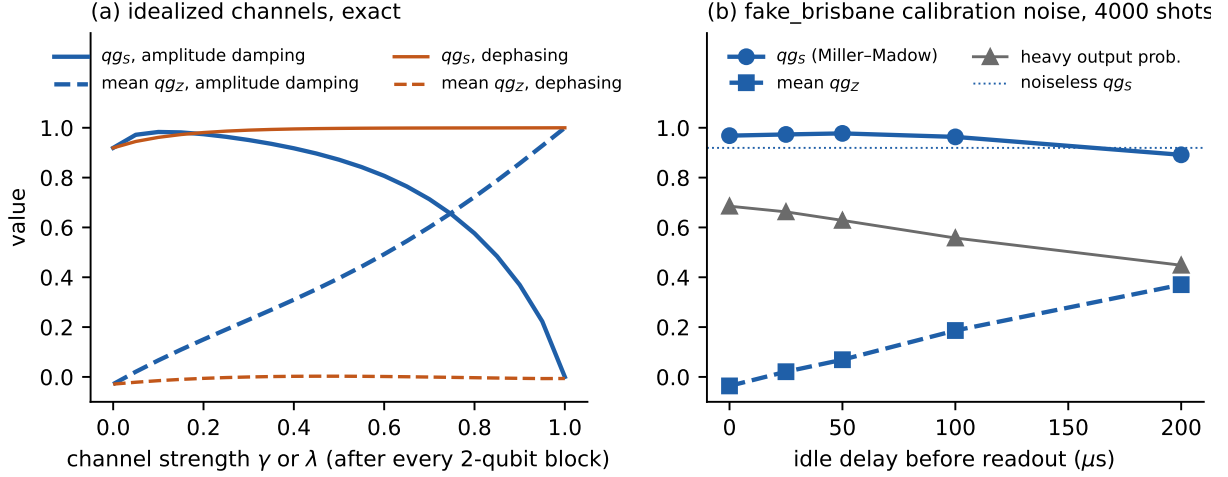


Figure 1: (a) 4-qubit Quantum Volume circuit, exact density-matrix simulation with amplitude damping or dephasing after every two-qubit block.  $qgs$  is non-monotonic under amplitude damping, while the mean  $qgz$  separates the regimes. (b) The same circuit under **fake\_brisbane** calibration noise with an idle delay before readout. Heavy output probability and mean  $qgz$  are monotonic;  $qgs$  falls below its noiseless value (dotted).

## 6 Further results (summary)

After the first draft, the same code base was used for twenty-six further studies. Each compares a qg-based choice with the standard one and with a classical control, and each is documented in **RESEARCH\_NOTES.md** (section numbers below) with a script and pinned tests. Table 3 lists them. The column “gain” states against what the qg choice wins, if at all. None of them is a quantum advantage over classical computation. Where a classical method was run, it matched or beat the quantum model at these sizes. The symmetry-witness studies (§20, §21, §24, §28–§33, §38, §39) are developed in a companion paper [16].

## 7 Limitations

(i) The Jacobian singularity at  $\theta \in \{0, \pi\}$ ; regularization costs exactness there. (ii) The arccos range: qg-space updates cannot reach  $\theta < 0$ . (iii) Z-basis blindness to phase-encoded entanglement and to readout dephasing. (iv)  $C_Z$  is not an entanglement measure. (v)  $qgs$  is not monotonic under T1. The pair  $(qgs, q\bar{q}z)$  fixes this empirically, in 23 of 24 circuits, not provably. (vi) Finite-shot bias of  $qgs$ . (vii) Pole damping costs iterations and has a trapping mode. (viii) All device results, here and in the follow-up studies, are calibration-based or vendor-model simulations. Runs on real hardware (`-mode ibm`, IonQ) are prepared but have not yet been performed.

## 8 Code availability

The library, every example, and the regression tests pinning every number above (859 tests at the time of this revision) are available at <https://github.com/Viny2030/qang> (MIT license). Figure 1 is regenerated by `manuscript/make_figures.py`. Derivations with additional detail are in **RESEARCH\_NOTES.md**.

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Generative AI tools (Anthropic Claude 3.5 Sonnet and Claude Opus 5.5; OpenAI GPT-4o) assisted with code generation and refactoring, test expansion, numerical experiments, and drafting of this text. The author framed the research questions, made the design decisions, and reviewed, edited and validated all AI-assisted output, including every derivation and number reported here.

## A Riesz–Fréchet and why $qg_Z$ is the canonical linear readout

Operators on  $\mathbb{C}^{2^n}$  form a Hilbert space under  $\langle A, B \rangle = \text{Tr}(A^\dagger B)$ . By the Riesz–Fréchet representation theorem [21, 22], every linear functional on it has the form  $\rho \mapsto \text{Tr}(O\rho)$  for a unique  $O$ . In finite dimension this is elementary linear algebra, so the appendix records a foundation rather than a new result. The functional  $\rho \mapsto qg_Z(\rho)$  is represented by  $\sigma_z$ , and  $q\bar{g}_Z$  by  $\frac{1}{n} \sum_i Z_i$ . Every linear readout, including linear XEB, has this form and hence an unbiased sample-mean estimator.  $qg_S$  is strictly concave in  $\rho$ , so no operator represents it, and its plug-in estimator is biased by Jensen’s inequality. By Proposition 1,  $qg_S$  is a fixed scalar function of the linear quantity  $qg_Z$ , so it carries no information about a single qubit beyond what  $\sigma_z$  extracts. (Riesz’s 1918 lemma on almost-orthogonal vectors concerns infinite-dimensional spaces and plays no role here.)

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Table 3: Follow-up studies (section numbers of RESEARCH\_NOTES.md).

| §      | topic                                       | main number                                                                                                               | gain                                                 |
|--------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 15     | natural gradient, cut cost, few-shot $qg_Z$ | QFI = $1/(1 - qg_Z^2)$ ; cut $\gamma = 1 + 2\sqrt{1 - qg_Z^2}$ ; Haar prior 92% vs 8% coverage near a pole                | closed forms; vs delta method                        |
| 16, 19 | QML encoding $\theta = \arccos x$           | Chebyshev models fit polynomials to $10^{-13}$ vs $5 \times 10^{-5}$ ; $\sim 10\times$ with shots                         | vs angle encoding [17]; classical Chebyshev fit wins |
| 17     | control grid uniform in $qg_Z$              | probability error $\pi/2$ smaller; distribution loading TVD lower in 43/45 cases                                          | vs $\theta$ grid; loses on fidelity near poles       |
| 18     | T1 vs unital noise from counts              | accuracy 0.88–0.91 vs 0.70–0.76 (XEB, HOP), strong noise only                                                             | vs XEB/HOP                                           |
| 20     | IonQ noise models                           | $H_2 + qg_Z$ filter $\sim 12$ vs $\sim 30$ mHa, 2–3% shots dropped                                                        | vs readout mitigation                                |
| 21     | $H_2$ electron-number witness               | 5.5 vs 18 mHa (HF 20.3); $11\times$ below HF at 2.5 Å; no gain on 6-qubit LiH                                             | vs readout mitigation, HF                            |
| 22     | ODE solver [18]                             | exponential decay up to $17\times$ better; not in general                                                                 | vs angle encoding; spectral method wins              |
| 23     | local cost $(1 - q\bar{g}_Z)/2$             | gradient variance $5600\times$ larger at 12 qubits; light cone trains 100 qubits classically                              | vs global cost [15]; classically simulable [19]      |
| 24     | $qg_Z$ filter vs ZNE by noise type          | witness predicts when the filter works; ZNE + filter 2.0 mHa on <b>fake_brisbane</b>                                      | complementary to ZNE                                 |
| 25     | thermal qubit, $qg_Z = \tanh \beta h$       | optimal thermometer $qg_Z \operatorname{artanh} qg_Z = 1$ [20]                                                            | none (prior, not qg)                                 |
| 26     | which gates to cut                          | exact optimum over 3795 partitions; $2\times$ fewer shots than counting                                                   | vs angle-blind rules; same as <b>find_cuts</b>       |
| 27     | repetition code choice under T1             | $qg_Z + qg_X$ policy regret 2.7% vs 46–258%                                                                               | decision rule                                        |
| 28     | spin-resolved filters                       | 23.2 $\rightarrow$ 14.7 mHa under depolarizing noise; none on <b>fake_brisbane</b>                                        | conditional                                          |
| 29     | 1D Hubbard, 8 qubits                        | ZNE + spin filters 6–9 $\times$ below raw; wall at $\geq 188$ ECR                                                         | vs raw, ZNE                                          |
| 30     | QAOA, exactly $K$                           | filter doubles $P(\text{opt})$ of XY-QAOA; penalty QAOA $\approx$ random guess                                            | vs QAOA; greedy wins                                 |
| 31     | qubit characterization                      | unbiased $T_{\text{eff}}$ ( $\pm 1$ mK) and readout error, where the standard suite reads 0 K                             | vs standard calibration                              |
| 32     | Leung code vs T1                            | infidelity $\approx 0.92\gamma^2$ vs $\gamma/3$ ; rule: Leung iff $T_2 > T_1$                                             | decision rule; code known                            |
| 33     | syndromes as witnesses                      | $\langle XXXX \rangle = qg_X^4$ ; drift tracking 13% regret with no probe shots                                           | vs periodic recalibration                            |
| 34     | entropy gap = coherence                     | qg sum bound captures 92–100% of Z coherence (thermal Ising)                                                              | none on graph states                                 |
| 35     | few-shot tomography                         | per-axis Haar prior best closed form on mixed states                                                                      | loses on pure states; prior, not qg                  |
| 36     | Ramsey operating point                      | $F = (V^2 - qg_Z^2)/(1 - qg_Z^2)$                                                                                         | none (known rule)                                    |
| 37     | Grover as $T_{2k+1}(qg_Z)$                  | $F_m = m^2/(1 - qg_Z^2)$ ; MLAE $\sim 9\times$ below Monte Carlo                                                          | none (known); noise must be modelled                 |
| 38     | IonQ noise models                           | filter doubles XY-QAOA; spin filters pay at depth                                                                         | vs QAOA; greedy wins                                 |
| 39     | native-gate ZNE on IonQ                     | CX folding is undone; at 2000 shots filter beats ZNE in RMS                                                               | depends on shot budget                               |
| 45     | shot noise, mixed states and registers      | $\text{Var}(\hat{\theta}) = [1 + (1 - r^2)/(r^2 \sin^2 \theta)]/N$ ; witness variance 0 on Dicke states, $n\times$ on GHZ | none (standard statistics); corrects §5 scope        |